New Mexico Quantum Technologies Award

Budget Narrative

SanTed Quantum Scheduling (SQS) · 2026 Program Year

Applicant: SanTed Quantum Scheduling (SQS)
Principal contact: Vernon T. Cox, DBA
Address: 2145 Isleta Blvd SW, Albuquerque, NM 87105-4643
EIN: 20-1529126
NM Business Tax ID (BTIN): 03-194279-00-2
Requested award amount: $200,000

Prior TIO funding

SQS is not currently receiving funding from the New Mexico Technology & Innovation Office. The NMQTA award would be SQS's first TIO engagement.

Overall scope of work

The 12-month grant period is structured around four quarterly milestones that move SQS from prototype stage to a pilot-validated, quantum-integrated scheduling MVP with commercial fleet operators. Every budget line item is tied directly to a milestone deliverable, and every deliverable is designed to produce tangible artifacts — signed pilot agreements, benchmark reports, patent filings, and documented ROI results — that can be reviewed and verified by TIO.

Budget summary

Row	Line item	Amount	% of award
7	Quantum algorithms engineer (12-month partial salary)	$40,000	20%
8	D-Wave Leap API access & quantum compute costs	$55,000	27.5%
9	Software infrastructure (cloud, DevOps, security audit)	$40,000	20%
10	Commercial client pilot program (Republic, Greyhound)	$30,000	15%
11	IP and patent filing (QUBO scheduling formulation)	$20,000	10%

Row	Line item	Amount	% of award
12	Conferences and trade shows (fleet / logistics industry)	$15,000	7.5%
16	**TOTAL**	**$200,000**	**100%**

Grant compliance
Payroll (Row 7) = 20% of award, within the 25% cap. Conferences (Row 12) = 7.5% of award, within the 10% cap. No funds allocated to alcohol, drugs, or food. All activities are New Mexico-based or directly support New Mexico activity.

Line item detail

Row 7 — Quantum algorithms engineer | $40,000 | 20% of award

Description

Partial-year salary for a quantum algorithms engineer hired in Q1, based in Albuquerque, New Mexico. The hire will formulate QUBO representations of fleet and operational scheduling problems, manage the D-Wave Leap API integration, tune penalty weights for Republic Services and Greyhound-style problem structures, and support benchmark testing.

Timeline

Recruitment and onboarding in Q1 (months 1–3); active development Q1–Q4. $40,000 represents approximately 9 months of engagement at a market-competitive rate for a quantum computing specialist in New Mexico, supplemented by SQS internal resources.

Justification

This is the single most critical hire in the grant period. Vernon Cox's domain expertise in scheduling is the company's foundation, but the QUBO formulation layer requires specialized quantum computing training. Recruiting this talent in New Mexico — leveraging UNM, NMSU, and the national laboratory community — directly supports NM economic development and keeps the hire within the state, satisfying NMQTA's in-state activity requirement.

Payroll compliance

$40,000 = 20% of the total award, within the grant's 25% payroll cap.

Milestone tie-in

Q1 — Engineer onboarded in NM; QUBO formulation spec v1.0 documented for fleet scheduling problems.

Row 8 — D-Wave Leap API access and quantum compute costs | $55,000 | 27.5% of award

Description

Commercial subscription and usage fees for D-Wave's Leap quantum cloud platform, which provides access to D-Wave's quantum annealing hardware and hybrid solver infrastructure. Includes annual subscription tier and per-job compute charges for development, benchmarking, and pilot execution.

Timeline

Subscription initiated in Q1 alongside engineer onboarding; active quantum compute usage ramps through Q2 benchmarking and Q3 pilot execution with Republic Services and Greyhound; final validation runs in Q4.

Justification

D-Wave Leap is the specific quantum hardware infrastructure required for SQS's product. As the largest line item, this spend directly signals to TIO that NMQTA funds are producing real quantum activity — not just classical software development. Without dedicated quantum compute budget, the core technology cannot be built.

Milestone tie-in

Q2 — D-Wave Leap integrated with SPS platform; benchmark report produced against classical routing tools on fleet-scale test data.

Row 9 — Software infrastructure | $40,000 | 20% of award

Description

Cloud hosting (AWS or Azure), DevOps and CI/CD tooling, development licenses, secure data handling infrastructure, and a third-party security audit to prepare the platform for commercial client engagement.

Timeline

Infrastructure buildout in Q1–Q2; security audit in Q3 ahead of pilot deployment with Republic Services and Greyhound.

Justification

Large commercial fleet operators handle sensitive operational data — routing, driver, and customer information — and require demonstrable security posture before allowing a new vendor to process their scheduling data. The security audit line item is non-negotiable for commercial pilot viability. Cloud infrastructure costs scale with the development workload and pilot data volume.

Milestone tie-in

Q2 — Production-grade infrastructure deployed. Q3 — Security audit complete, enabling commercial pilot execution.

Row 10 — Commercial client pilot program | $30,000 | 15% of award

Description

Direct costs associated with executing a structured pilot with Republic Services and Greyhound Lines, including business development outreach, travel to client operations sites, data preparation and anonymization, pilot coordination, and documentation of benchmark results.

Timeline

Outreach and pilot scoping in Q2; agreement negotiation late Q2; active pilot execution in Q3; final results documentation in Q4.

Justification

Without documented pilot results from real commercial fleet operators, the quantum engine is an unvalidated engineering exercise. Pilot execution costs turn the technology into commercial evidence — benchmark reports, ROI data, and customer testimonials — that support commercial sales to comparable fleet operators in year two.

Milestone tie-in

Q3 — Signed pilot agreements with Republic Services and Greyhound, interim results. Q4 — Final pilot results with documented ROI published.

Row 11 — IP and patent filing | $20,000 | 10% of award

Description

Legal and filing fees for a provisional patent application covering SQS's QUBO scheduling formulation method — specifically, the proprietary approach to encoding complex fleet and operational scheduling constraints as quantum-ready binary optimization problems.

Timeline

Patent attorney engagement in Q3; provisional application filed in Q4.

Justification

The QUBO formulation is SQS's core defensible IP — the layer that combines commercial scheduling domain expertise with quantum computing. Filing a provisional patent establishes priority date, protects the company's competitive position, and materially strengthens SQS's positioning for SBIR Phase II and Series A fundraising in year two.

Milestone tie-in

Q4 — Provisional patent application filed.

Row 12 — Conferences and trade shows | $15,000 | 7.5% of award

Description

Registration, travel, and exhibit fees for SQS participation in two targeted industry events — Waste Expo (waste and recycling fleet industry), the American Trucking Associations Management Conference, or equivalent commercial fleet and logistics industry events.

Timeline

Conference attendance staged in Q3 and Q4 to coincide with pilot result availability and commercial sales conversations.

Justification

Commercial fleet and logistics industry events concentrate decision-makers from Republic Services-comparable operators in one place. A single commercial engagement captured through conference presence would represent multiples of this line item's cost. Industry events are the fastest path from pilot results to next-customer pipeline.

Conference compliance

$15,000 = 7.5% of the total award, within the grant's 10% conference cap.

Milestone tie-in

Q4 — Pilot results presented; commercial sales pipeline initiated.

Budget total and compliance summary

Total requested: $200,000. Payroll: 20% (cap: 25%). Conferences: 7.5% (cap: 10%). No funds allocated to alcohol, drugs, or food. All budgeted activities take place within New Mexico or directly support New Mexico-based activities and commercial pilot engagements. Every dollar of NMQTA funding is tied to a specific Q1–Q4 milestone deliverable, enabling TIO to measure progress against commitments throughout the grant period.

Certification

SanTed Quantum Scheduling (SQS) certifies that the budget presented in this narrative is accurate, that all funds will be used for the purposes described, and that all expenditures will be documented with receipts and supporting records available for TIO review upon request throughout the grant period and in the two-year post-award reporting period.

Signed: ______________________________

Name: Vernon T. Cox, DBA
Title: Founder & Principal, SanTed Quantum Scheduling
Date: ___________________________